Problem 1: Let V be a finite dimensional vector space. Suppose $\alpha = (u_1, \ldots, u_n)$ and $\beta = (v_1, \ldots, v_n)$ be ordered bases of V. Let $T: V \to V$ be a linear operator be s.t. $Tv_j = u_j$. Then prove that
$$[T]_\beta^\beta = [I]_\alpha^\beta.$$

Proof: The j^{th} column of $[T]_\beta^\beta$ is the column representation of $[Tv_j]^\beta$ and j^{th} column of

$[I]_\alpha^\beta$ is $[Iu_j]^\beta$.

$$[Tv_j]^\beta = [u_j]^\beta \quad \text{by definition of } T.$$

$$[Iu_j]^\beta = [u_j]^\beta = [Tv_j]^\beta.$$

Hence the j^{th} column of $[T]_\beta^\beta$ is the same as the j^{th} column of $[I]_\alpha^\beta$.

Since this true for every column.

$$[T]_\beta^\beta = [I]_\alpha^\beta .$$

Problem: Let $T: V \longrightarrow W$ be a function.
Then the graph of T is the subset

$$\text{graph}(T) := \{ (v, Tv) \in V \times W : v \in V \}.$$

Prove that graph(T) is a subspace of $V \times W$ iff T is a linear transformation.

Proof: Assume that graph(T) is a subspace of $V \times W$.

Let $v_1, v_2 \in V$ and $c \in \mathbb{R}$. Want to check that

$$T(v_1 + c v_2) = Tv_1 + c\,Tv_2.$$

Since graph(T) is a subspace and $(v_1, Tv_1) \in$ graph(T), $(v_2, Tv_2) \in$ graph(T)

we have

$$(v_1, Tv_1) + c(v_2, Tv_2) \in \text{graph}(T)$$

i.e. $(v_1 + c v_2, \; Tv_1 + c\,Tv_2) \in \text{graph}(T)$

By the defn of graph(T), $\exists$ $w \in V$ s.t

$$(v_1 + cv_2, \; Tv_1 + cTv_2) = (w, Tw).$$

$$\Rightarrow \quad v_1 + cv_2 = w \longrightarrow (*)$$

$$Tv_1 + cTv_2 = Tw$$

But $\Rightarrow$ $Tv_1 + cTv_2 = T(v_1 + cv_2)$ by $(*)$.

Conversely, Let (v_1, Tv_1) and $(v_2, Tv_2) \in$ graph(T).

Then $(v_1, Tv_1) + (v_2, Tv_2) = (v_1 + v_2, \; Tv_1 + Tv_2)$

$$= \left(v_1 + v_2 ,\ T(v_1+v_2) \right)$$ since T - linear transformation.

$$\in \operatorname{graph}(T)$$

Hence graph(T) is closed under vector addition.

Similarily graph(T) " " " scalar multiplication.

Hence graph(T) is a subspace.

<u>Problem</u>: Let $V_1, \ldots, V_m$ and W be vector spaces. Then prove that $\mathcal{L}(V_1 \times \ldots \times V_m, W)$ is isomorphic

to the vector space $\mathcal{L}(V_1, W) \times \mathcal{L}(V_2, W) \times \cdots \times \mathcal{L}(V_m, W)$.

Solution: If $V_1, \ldots, V_m$ and W are finite dimensional

Let $\dim(V_i) = n_i$ and $\dim(W) = k$.

$$\dim(V_1 \times \cdots \times V_m) = n_1 + n_2 + \cdots + n_m$$

& $$\dim(\mathcal{L}(V_1 \times \cdots \times V_m, W)) = (n_1 + \cdots + n_m)k$$

$$\dim(\mathcal{L}(V_j, W)) = n_j k$$

$$\dim(\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)) = n_1 k + n_2 k + \cdots + n_m k.$$

$$= \dim(\mathcal{L}(V_1 \times \cdots \times V_m, W))$$

Hence they are isomorphic.

Define $\Phi : \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W) \longrightarrow \mathcal{L}(V_1 \times \cdots \times V_m, W)$

$$\Phi(T_1, \ldots, T_m) = T$$

where $T(v_1, \ldots, v_m) := T_1 v_1 + T_2 v_2 + \cdots + T_m v_m.$

$$T\big((v_1, \ldots, v_m) + (u_1, \ldots, u_m)\big) = T(v_1 + u_1, \ldots, v_m + u_m)$$

$$= T_1(v_1 + u_1) + \cdots + T_m(v_m + u_m)$$

$$= T_1 v_1 + T_1 u_1 + \cdots + T_m v_m + T_m u_m$$

$$= (T_1 v_1 + \cdots + T_m v_m) + (T_1 u_1 + \cdots + T_m u_m)$$

$$= T(v_1, \ldots, v_m) + T(u_1, \ldots, u_m)$$

$|||^{ly}$ $\quad T(c(v_1, \ldots, v_m)) = c\, T(v_1, \ldots, v_m)$

<u>Claim:</u> Φ is a linear transformation.

Let $(T_1, \ldots, T_m)$ & $(S_1, \ldots, S_m) \in \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$.

$$\Phi\big((T_1, \ldots, T_m) + (S_1, \ldots, S_m)\big) = \Phi\big((T_1 + S_1, T_2 + S_2, \ldots, T_m + S_m)\big)$$

$$\Phi(T_1+S_1,\ldots,T_m+S_m)(v_1,\ldots,v_m) =$$

$$(T_1+S_1)v_1 + \cdots + (T_m+S_m)v_m$$

$$= T_1v_1 + S_1v_1 + \cdots + T_mv_m + S_mv_m$$

$$= (T_1v_1 + \cdots + T_mv_m) + (S_1v_1 + \cdots + S_mv_m)$$

$$= \Phi(T_1,\ldots,T_m)(v_1,\ldots,v_m) + \Phi(S_1,\ldots,S_m)(v_1,\ldots,v_m)$$

$$= \Big(\Phi(T_1,\ldots,T_m) + \Phi(S_1,\ldots,S_m)\Big)(v_1,\ldots,v_m)$$

$$\forall \; (v_1,\ldots,v_m) \in V_1 \times \cdots \times V_m$$

$$\Rightarrow \quad \Phi\big((T_1,\ldots,T_m)+(S_1,\ldots,S_m)\big) = \Phi(T_1,\ldots,T_m) + \Phi(S_1,\ldots,S_m)$$

III^{ly} $\Phi(c(T_1, \ldots, T_m)) = c\,\Phi(T_1, \ldots, T_m)$

Hence Φ is a linear transformation.

<u>Claim:</u> Φ is bijective.

<u>Injectivity:</u> We shall show that $\mathrm{Null}(\Phi) = \{0\}$.

Let $(T_1, \ldots, T_m) \in \mathrm{Null}(\Phi)$

$\Rightarrow$ $\Phi(T_1, \ldots, T_m) = 0$ (i.e. $0 : V_1 \times \cdots \times V_m \longrightarrow W$).

i.e for any $v_1 \in V_1$

$$\Phi(T_1, \dots, T_m)(v_1, 0, \dots, 0) = 0$$

$$\Rightarrow \quad T_1 v_1 + 0 + \quad + 0 = 0$$

$$\Rightarrow \quad T_1 v_1 = 0$$

$$\Rightarrow \quad T_1 \equiv 0$$

|||ly $T_j \equiv 0 \quad \forall \ 1 \le j \le m.$

Therefore $\text{null}(\Phi) = \{0\}.$

Surjectivity

Let $T \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$

Define $T_i \in \mathcal{L}(V_i, W)$ to be

$$T_i v_i = T(0, \ldots, 0, \underset{\substack{\uparrow \\ i\text{th coordinate}}}{v_i}, 0, \ldots, 0)$$

Check that T_i is a linear transformation.

$$\Phi(T_1, \ldots, T_m)(v_1, \ldots, v_m) = T_1 v_1 + \cdots + T_m v_m$$

$$= T(v_1, 0, \ldots, 0) + T(0, v_2, 0 \ldots, 0) + \cdots + T(0, \ldots, 0, v_m)$$

$$= T(v_1, v_2, \ldots, v_m)$$

Hence Φ is surjective

$\Rightarrow$ Φ is an isomorphism.

Problem 4: Let V be a vector space. Let $v, v' \in V$ and U, W be subspaces of V s.t $v + U = v' + W$. Then prove that $U = W$.

Proof: Since $v + \mathcal{U} = v' + W$

$\exists$ $w \in W$ s.t

$$v + 0 = v' + w$$

$$\Rightarrow \quad (v - v') = w \in W.$$

Let $u \in \mathcal{U}$. Enough to show that $u \in W$.

$\exists$ $w' \in W$ s.t

$$v + u = v' + w'$$

$$\Rightarrow \quad u = -(v - v') + w'$$

$$\in W$$

$\Rightarrow \quad \mathcal{U} \subseteq W$

$|||^{ly}$ $\quad W \subseteq \mathcal{U}$ and hence $\mathcal{U} = W$.

<u>Problem</u>: Let V be a finite dimensional vector space and U be a subspace of V. Let $(v_1 + \mathcal{U}, \ldots, v_n + \mathcal{U})$ be a basis of $V/\mathcal{U}$. Further let $(u_1, \ldots, u_m)$ be an ordered basis of U. Then $(v_1, \ldots, v_n, u_1, \ldots, u_m)$ is an ordered basis of V.

Proof: We know that

$$\dim(V/U) = \dim(V) - \dim(U)$$

$$\Rightarrow \dim(V) = \dim(V/U) + \dim(U).$$

$$\Rightarrow \dim(V) = n + m.$$

Consider $\beta = (v_1, \ldots, v_n, u_1, \ldots, u_m)$.

Enough to show that β is linearly independent.

Let $a_1, \dots, a_n, b_1, \dots, b_m \in \mathbb{R}$ be s.t.

$$a_1 v_1 + \cdots + a_n v_n + b_1 u_1 + \cdots + b_m u_m = 0$$

Recall that $\pi : V \to V/U$ given by

$\pi(v) = v + U$ is a linear transformation.

Thus $\pi(a_1 v_1 + \cdots + a_n v_n + b_1 u_1 + \cdots + b_m u_m) = 0$

$$\Rightarrow a_1(v_1 + U) + \cdots + a_n(v_n + U) + b_1 \underbrace{\cancel{(u_1 + U)}}_{0} + \cdots + b_m \underbrace{\cancel{(u_m + U)}}_{0} = 0$$

$$\Rightarrow \quad a_1(v_1 + U) + \cdots + a_n(v_n + U) = 0$$

$$\Rightarrow \quad a_1 = a_2 = \cdots = a_n = 0 \quad \left(\because \ (v_1 + U, \ldots, v_n + U) \text{ is a basis of } V/U \right).$$

$$\Rightarrow \quad b_1 u_1 + \cdots + b_m u_m = 0$$

$$\Rightarrow \quad b_1 = b_2 = \cdots = b_m = 0 \quad \left(\because \ (u_1, \ldots, u_m) \text{ is a basis of } U \right).$$

$$\therefore \quad a_i = 0 \ \forall i \qquad b_j = 0 \ \forall j$$

$\Rightarrow$ $(v_1, \ldots, v_n, u_1, \ldots, u_m)$ is linearly dependent.

$\Rightarrow \quad (v_1, \ldots, v_n, u_1, \ldots, u_m)$ is a basis of V.

Problem: Let $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be defined by

$$T(p) = x^2 p(x) + p''(x).$$

(i) Suppose $\varphi \in \mathcal{P}(\mathbb{R})^*$ is defined by $\varphi(p) = p'(4)$. Then describe the linear functional $T^t\varphi$ on $\mathcal{P}(\mathbb{R})$

(ii) Suppose $\varphi \in \mathcal{P}(\mathbb{R})^*$ is defined by $\varphi(p) = \int_0^1 p(x)\,dx$. Then evaluate $(T^t\varphi)(x^3)$

Solution: (i) Recall that $T^t\varphi = \varphi T$

$$(T^t\varphi)(p(x)) = \varphi(T(p(x))).$$

$$= \varphi(x^2 p(x) + p''(x)).$$

$$= \left(2x\,p(x) + x^2 p'(x) + p'''(x)\right)\Big|_4$$

$$= 8p(4) + 16p'(4) + p'''(4).$$

(ii) $(T^t\varphi)(x^3) = \varphi(T(x^3)).$

$$= \varphi\left(x^5 + 6x \right)$$

$$= \int_0^1 (x^5 + 6x)\, dx$$

$$= \left(\frac{x^6}{6} + 3x^2 \right)\Big|_0^1 = \frac{1}{6} + 3 = \frac{19}{6}$$